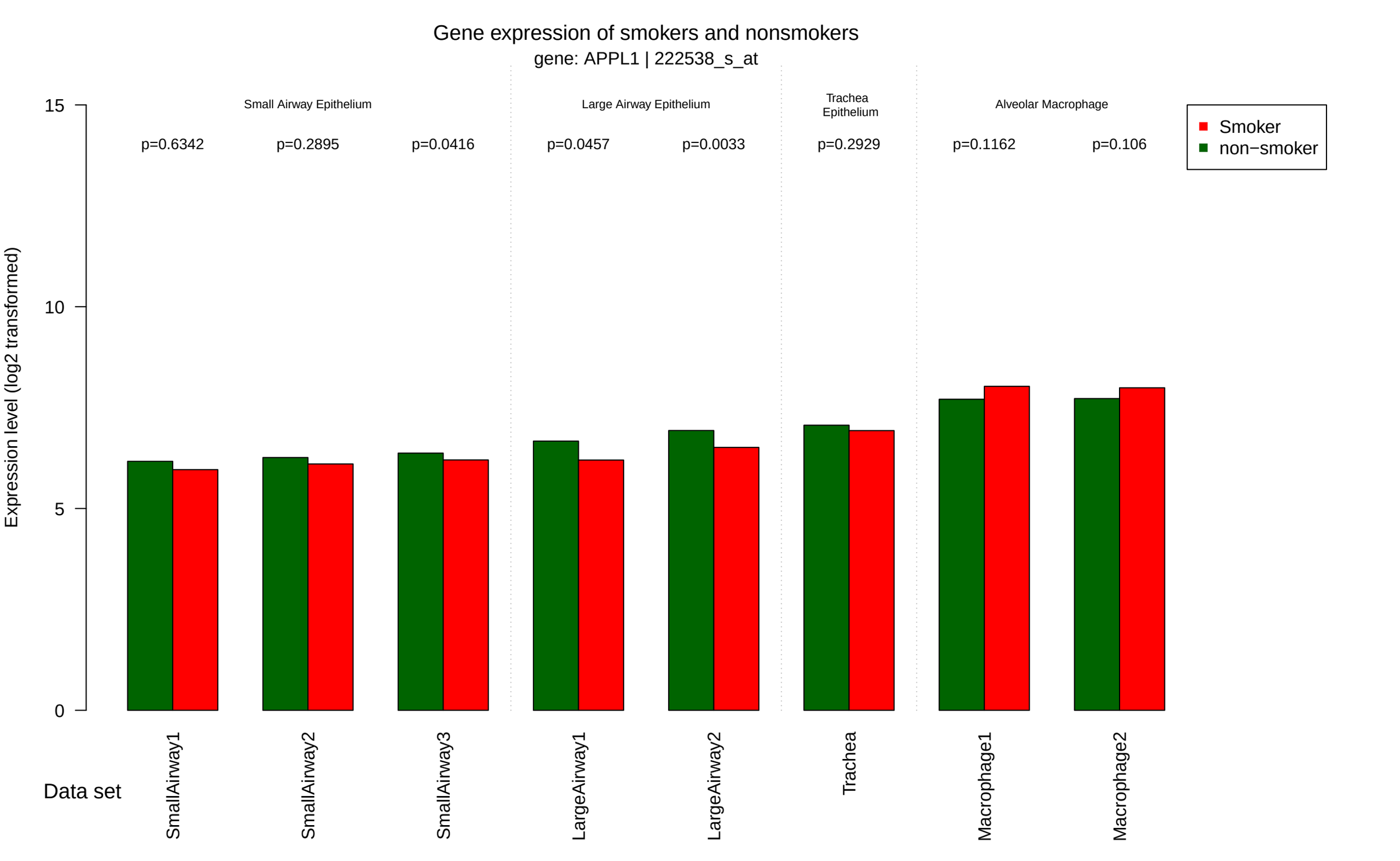
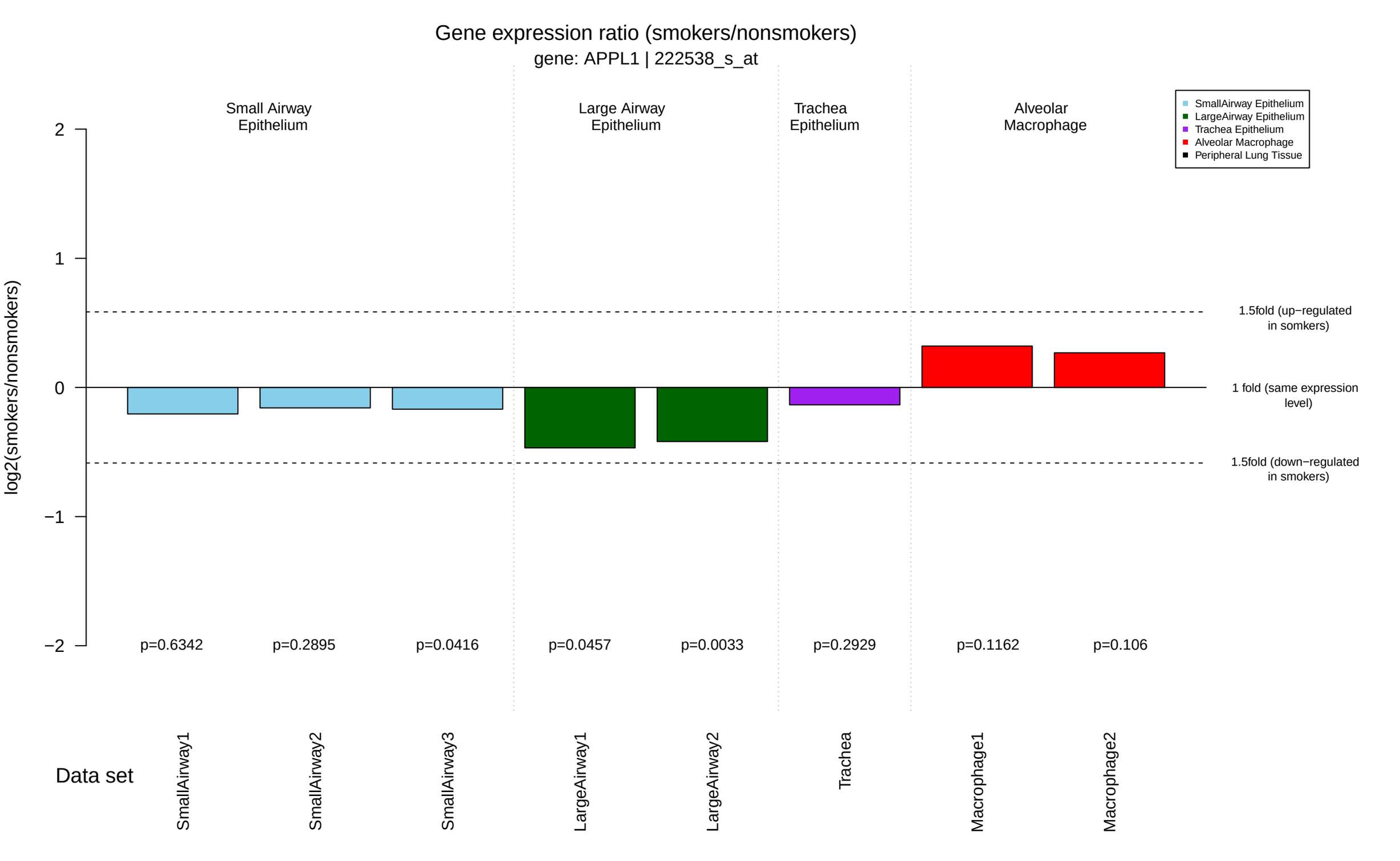
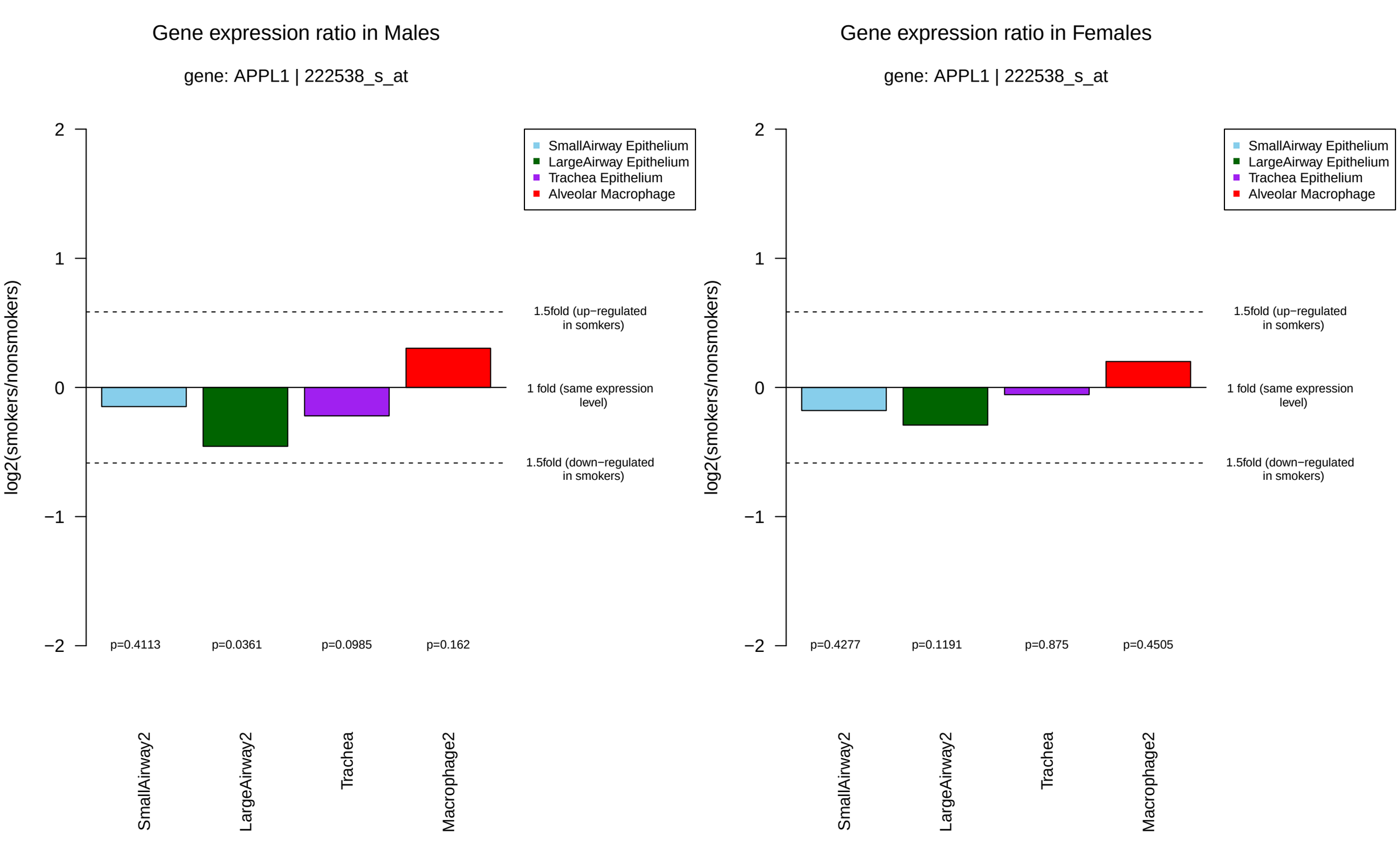
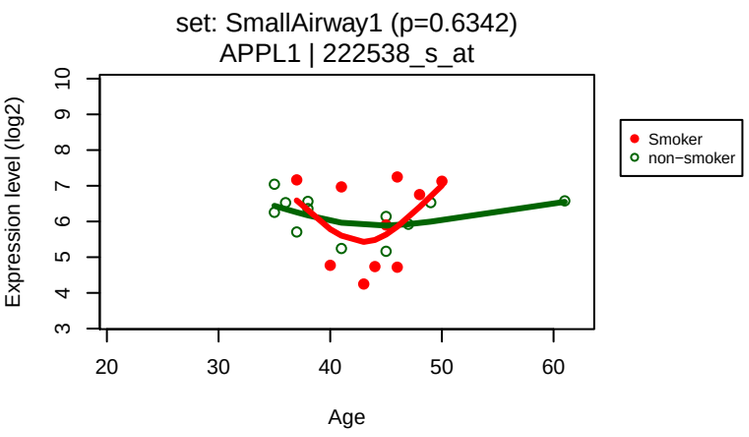
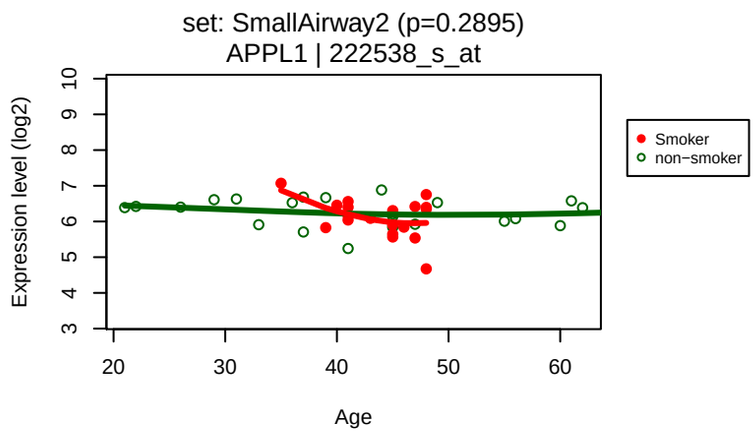
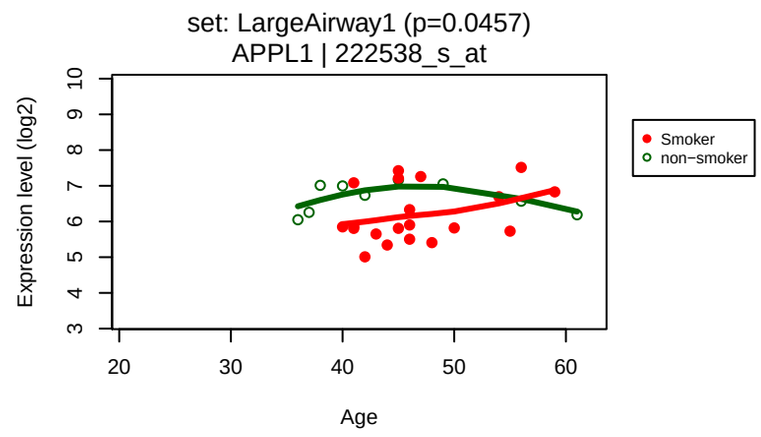

Gene expression of smokers and nonsmokers
gene: APPL1 | 222538_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6342
p=0.2895
p=0.0416
p=0.0457
p=0.0033
p=0.2929
p=0.1162
p=0.106
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

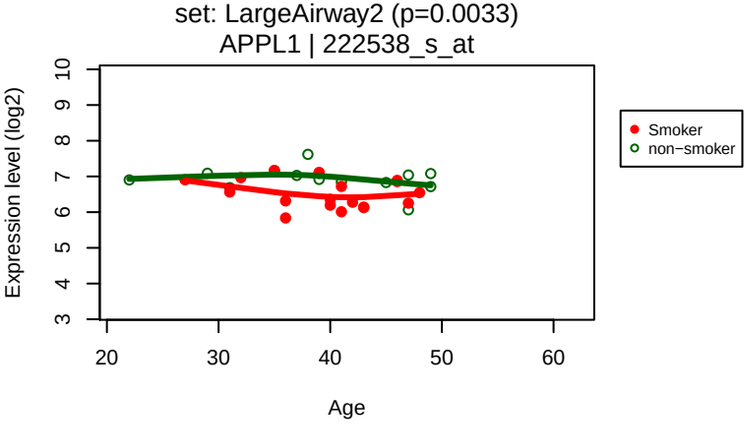

Gene expression ratio (smokers/nonsmokers)
gene: APPL1 | 222538_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6342
p=0.2895
p=0.0416
p=0.0457
p=0.0033
p=0.2929
p=0.1162
p=0.106
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

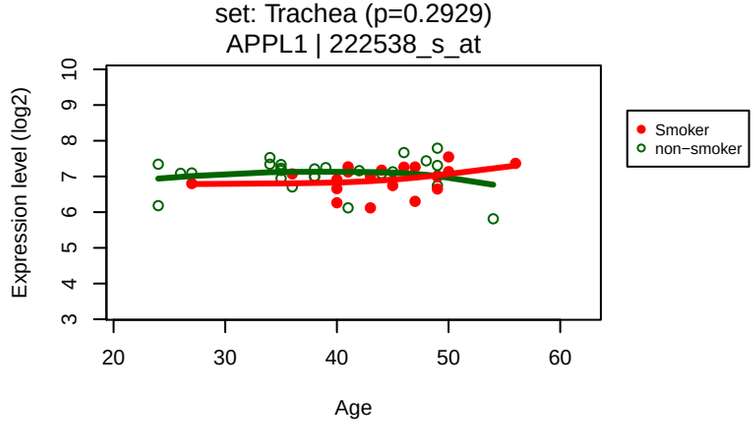

Gene expression ratio in Males
gene: APPL1 | 222538_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4113
p=0.0361
p=0.0985
p=0.162
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: APPL1 | 222538_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4277
p=0.1191
p=0.875
p=0.4505
SmallAirway2
LargeAirway2
Trachea
Macrophage2

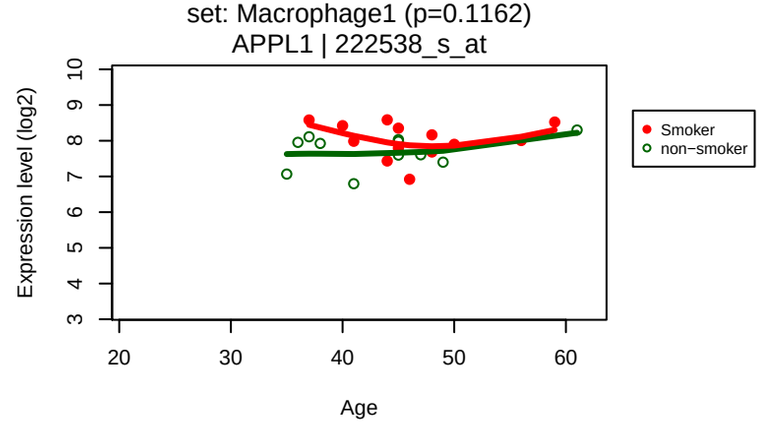

set: SmallAirway1 (p=0.6342)
APPL1 | 222538_s_at
Expression level (log2)
Age
Smoker
non−smoker

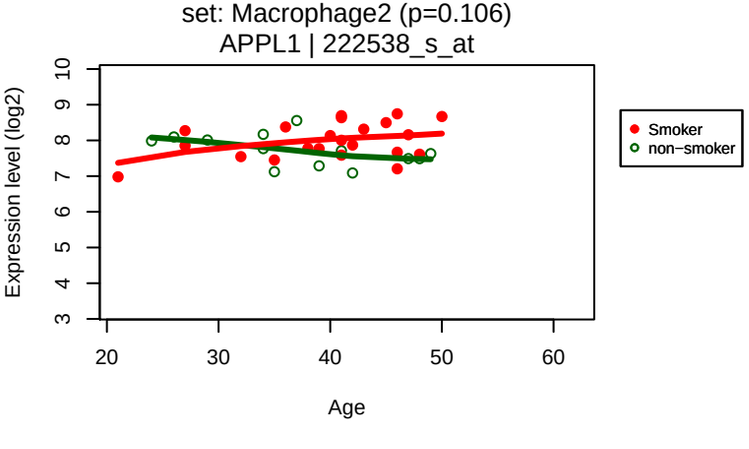

set: SmallAirway2 (p=0.2895)
APPL1 | 222538_s_at
Expression level (log2)
Age
Smoker
non−smoker

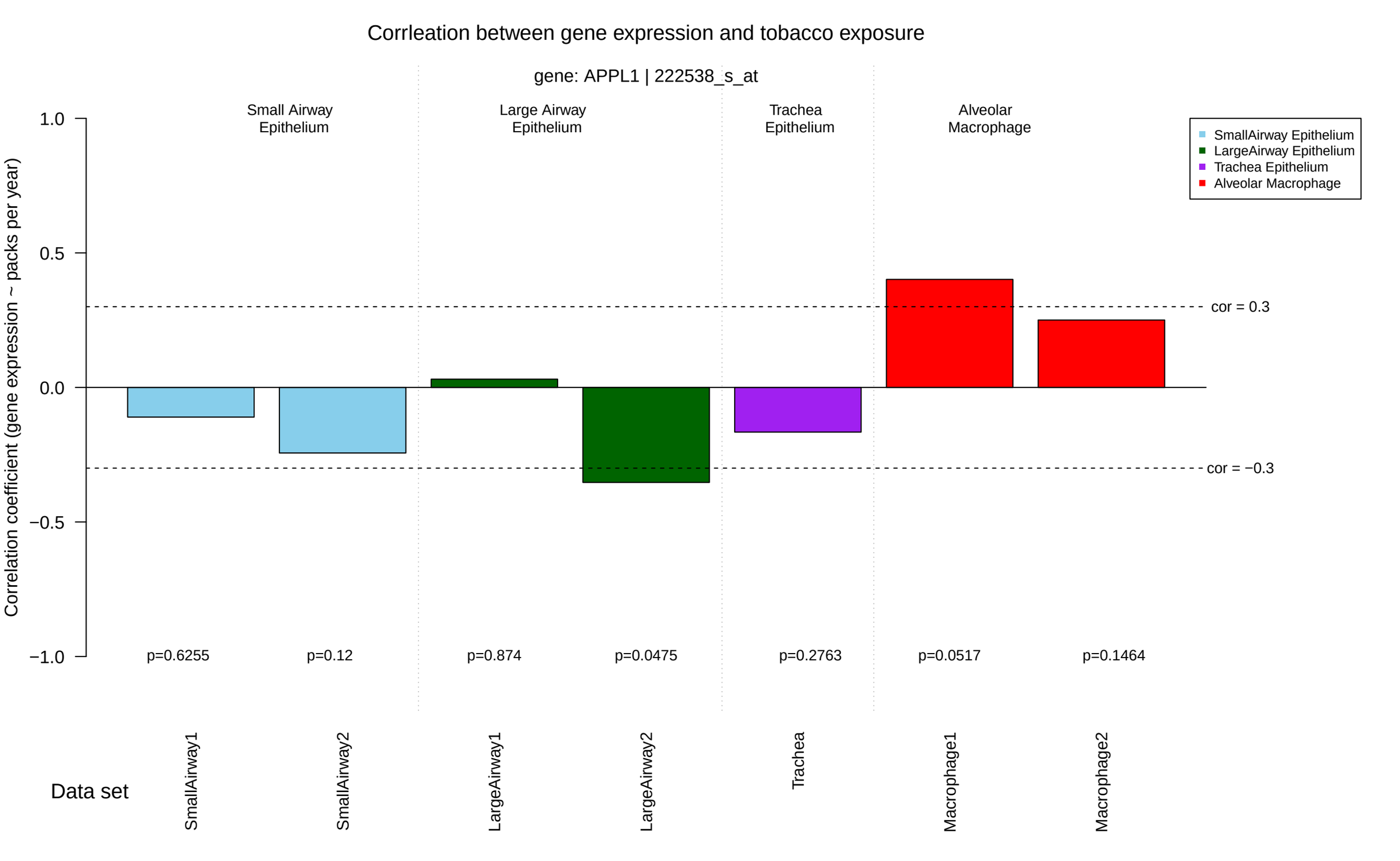

set: LargeAirway1 (p=0.0457)
APPL1 | 222538_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0033)
APPL1 | 222538_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2929)
APPL1 | 222538_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1162)
APPL1 | 222538_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.106)
APPL1 | 222538_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: APPL1 | 222538_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6255
p=0.12
p=0.874
p=0.0475
p=0.2763
p=0.0517
p=0.1464
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2